# Nonlinear Lattice Boltzmann Formulation for Analog Video Synthesis and Spinodal-Like Pattern Formation

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[1] *Computational Physics and Nonlinear Dynamics Research*

(Dated: November 12, 2026)

Analog and hybrid video feedback systems generate complex, self-organizing lamellar and labyrinthine patterns that bear a strong morphological resemblance to spinodal decomposition in quenched binary fluid mixtures. In this work, we establish a theoretical and computational bridge between Cahn-Hilliard phase-separation dynamics and discrete-time optical-electronic video feedback loops. We formulate a two-dimensional, nine-velocity (D2Q9) lattice Boltzmann model incorporating a sigmoidal nonlinear collision operator to emulate analog mixer saturation and optical point-spread diffusion. Furthermore, we present a modular, compile-time polymorphic C++ software architecture designed for zero-overhead extensibility in real-time video synthesis applications. Preliminary numerical simulations initialized with stochastic density perturbations demonstrate the transition from uncorrelated high-frequency noise to coherent, coarsening spatial domains governed by relaxation kinetics and nonlinear feedback gain.

## I. INTRODUCTION

Spontaneous pattern formation in far-from-equilibrium systems is a central theme across condensed matter physics, fluid dynamics, and nonlinear optics. In multiphase fluid dynamics, spinodal decomposition occurs when a homogeneous binary mixture is rapidly quenched into a thermodynamically unstable region of its phase diagram [1]. Driven by uphill diffusion, infinitesimal compositional fluctuations grow exponentially at a characteristic wavelength, yielding interconnected, bicontinuous domains that coarsen over time.

Remarkably similar morphologies emerge in analog and hybrid video synthesis when an imaging sensor captures the output of a cathode-ray tube (CRT) display routed through a nonlinear video mixer. In such optical-electronic feedback loops, spatial defocusing, phosphor persistence, and electronic voltage saturation interact to amplify specific spatial frequencies while suppressing others. Although practitioners in experimental video synthesis frequently invoke the terminology of spinodal decomposition to describe these striated, finger-like instabilities, a rigorous kinetic formulation mapping video synthesis signal paths onto mesoscopic fluid solvers remains largely unexplored.

The lattice Boltzmann method (LBM) provides an ideal mesoscopic framework for bridging these domains [2, 3]. Rather than solving macroscopic partial differential equations directly, LBM evolves discrete particle distribution functions on a regular lattice through localized collision and streaming operations. By modifying the collision operator and external forcing terms, LBM naturally accommodates non-ideal equations of state, interfacial gradient penalties, and custom nonlinear transfer functions.

In this paper, we formalize the mathematical correspondence between Cahn-Hilliard spinodal dynamics and video feedback loops, detail a nonlinear D2Q9 LBM formulation tailored for video synthesis, and outline a policy-based C++ architecture engineered for real-time execution.

## II. THEORETICAL FORMULATION

### A. Spinodal Decomposition vs. Video Feedback

In classical continuum thermodynamics, the evolution of a conserved order parameter $c(\boldsymbol{x}, t)$ in a binary mixture is governed by the Cahn-Hilliard equation:

$$\frac{\partial c}{\partial t} = \nabla \cdot \left[ M \nabla \left( \frac{\partial f(c)}{\partial c} - \kappa \nabla^2 c \right) \right], \tag{1}$$

where $M$ is the mobility, $f(c)$ is the bulk Helmholtz free-energy density, and $\kappa$ is the interfacial gradient energy coefficient penalizing sharp compositional transitions. When quenched into the spinodal region where $f''(c_0) < 0$, linear stability analysis predicts a fastest-growing unstable mode with characteristic wavelength

$$\lambda_m = 2\pi \sqrt{\frac{-2\kappa}{f''(c_0)}}. \tag{2}$$

In an optical-electronic video feedback system, a two-dimensional luminance or color intensity field $I(\boldsymbol{x}, t)$ evolves across discrete frame intervals $\Delta t$ according to the map:

$$I(\boldsymbol{x}, t + \Delta t) = \mathcal{N}\left( (\mathcal{K} * I)(\boldsymbol{x}, t) + \eta(\boldsymbol{x}, t) \right), \tag{3}$$

where $\mathcal{K}(\boldsymbol{x})$ denotes the spatial point-spread function (PSF) arising from camera lens defocus, sensor pixel pitch integration, and CRT phosphor persistence; $*$ denotes spatial convolution; $\eta(\boldsymbol{x}, t)$ represents external video inputs or stochastic thermal noise; and $\mathcal{N}(\cdot)$ is a smooth sigmoidal transfer function modeling amplifier gain and voltage-rail saturation.

TABLE I. Discrete velocity vectors $\boldsymbol{c}_i = (c_{ix}, c_{iy})$ and isotropic lattice weights $w_i$ for the D2Q9 model.

| Index $i$ | Velocity Vector $\boldsymbol{c}_i$ | Lattice Weight $w_i$ |
|---|---|---|
| 0 | $(0,0)$ | $4/9$ |
| $1,2,3,4$ | $(\pm1,0),(0,\pm1)$ | $1/9$ |
| $5,6,7,8$ | $(\pm1,\pm1)$ | $1/36$ |

Expanding the convolution $(\mathcal{K} * I)$ for an isotropic Gaussian PSF of width $\sigma$ to second order in spatial gradients yields a diffusive smoothing operator:

$$(\mathcal{K} * I)(\boldsymbol{x}, t) \approx I(\boldsymbol{x}, t) + \frac{\sigma^2}{2}\nabla^2 I(\boldsymbol{x}, t), \tag{4}$$

which plays a stabilizing, dissipative role analogous to the interfacial gradient penalty $\kappa\nabla^2 c$ in Eq. (1). Simultaneously, high-gain sigmoidal amplification $\mathcal{N}$ destabilizes uniform gray levels, driving intensity values toward upper and lower saturation bounds while generating harmonic spatial ripples.

### B. Nonlinear D2Q9 Lattice Boltzmann Model

To simulate these coupled diffusive and nonlinear amplification dynamics on a two-dimensional Cartesian grid, we employ the isotropic D2Q9 lattice velocity set. The discrete velocities $\boldsymbol{c}_i$ $(i = 0, \ldots, 8)$ and their corresponding lattice weights $w_i$ are summarized in Table I.

Macroscopic pixel density $\rho(\boldsymbol{x}, t)$ and advective velocity $\boldsymbol{u}(\boldsymbol{x}, t)$ are recovered from the moments of the single-particle distribution functions $f_i(\boldsymbol{x}, t)$:

$$\rho = \sum_{i=0}^{8} f_i, \qquad \rho\boldsymbol{u} = \sum_{i=0}^{8} f_i \boldsymbol{c}_i. \tag{5}$$

The local Maxwell-Boltzmann equilibrium distribution $f_i^{\mathrm{eq}}$ is given by

$$f_i^{\mathrm{eq}}(\rho, \boldsymbol{u}) = w_i \rho \left[1 + \frac{\boldsymbol{c}_i \cdot \boldsymbol{u}}{c_s^2} + \frac{(\boldsymbol{c}_i \cdot \boldsymbol{u})^2}{2c_s^4} - \frac{\boldsymbol{u} \cdot \boldsymbol{u}}{2c_s^2}\right], \tag{6}$$

where $c_s^2 = 1/3$ is the squared lattice speed of sound in lattice units ($\Delta x = \Delta t = 1$).

In standard Bhatnagar-Gross-Krook (BGK) kinetics, populations relax toward local equilibrium at rate $\omega = 1/\tau$. To emulate analog video mixer saturation within the mesoscopic collision step, we define a nonlinear post-collision state $f_i^\star$:

$$f_i^\star(\boldsymbol{x}, t) = L_{\mathrm{sat}} \tanh\left(\frac{g_{\mathrm{fb}}}{L_{\mathrm{sat}}}\Big[f_i(\boldsymbol{x}, t) - \omega\left(f_i(\boldsymbol{x}, t) - f_i^{\mathrm{eq}}(\boldsymbol{x}, t)\right)\Big]\right), \tag{7}$$

TABLE II. Simulation parameters for baseline slow-evolution video synthesis runs.

| Parameter | Symbol | Value |
|---|---|---|
| Grid resolution | $N_x \times N_y$ | $256 \times 256$ |
| Relaxation time | $\tau$ | 0.85 |
| Relaxation frequency | $\omega$ | 1.176 |
| Feedback gain | $g_{\mathrm{fb}}$ | 1.05 |
| Saturation limit | $L_{\mathrm{sat}}$ | 1.50 |
| Initial perturbation amplitude | $\delta\rho$ | 0.03 |
| Total time steps | $N_{\mathrm{steps}}$ | 400 |

where $g_{\mathrm{fb}}$ is the dimensionless feedback gain and $L_{\mathrm{sat}}$ sets the asymptotic saturation ceiling. Following collision and optional external forcing $F_i$, populations propagate to neighboring nodes via periodic pull-streaming:

$$f_i(\boldsymbol{x} + \boldsymbol{c}_i\Delta t, t + \Delta t) = f_i^\star(\boldsymbol{x}, t) + F_i(\boldsymbol{x}, t). \tag{8}$$

## III. COMPUTATIONAL ARCHITECTURE

Real-time video synthesis requires executing millions of lattice updates per second to maintain interactive frame rates ($\geq 60$ Hz). Traditional object-oriented plugin architectures rely on runtime polymorphism via virtual function tables, which inhibits compiler inlining and introduces branch prediction penalties inside tight $O(N_x N_y)$ spatial loops.

To achieve modularity without sacrificing computational throughput, we structure the core simulation engine (`VideoSynthLBM`) using compile-time policy-based template design in C++17. The core grid solver is parameterized over independent policy classes governing collision kinetics and external forcing:

```
LatticeGrid<CollisionPolicy, ForcingPolicy>.
```
(9)

During compilation, the concrete implementations of `CollisionPolicy::apply` and `ForcingPolicy::apply` are statically resolved and inlined directly into the primary time-stepping kernel.

Furthermore, to bridge high-performance C++ execution with rapid visualization and user interaction, the engine is exposed to Python via custom `pybind11` bindings. The lattice density field is mapped directly to zero-copy NumPy array views, enabling a lightweight Pygame graphical user interface to render real-time simulations at 60 Hz while dynamically adjusting kinetic parameters (such as relaxation time $\tau$ and feedback gain $g_{\mathrm{fb}}$) via keyboard controls and interactive UI sliders.

## IV. PRELIMINARY NUMERICAL RESULTS

Table II details the kinetic parameters selected to resolve early-stage coarsening. Setting the relaxation time

to $\tau = 0.85$ moderates viscous damping and diffusive transport, while a gentle feedback gain ($g_{\text{fb}} = 1.05$) prevents immediate numerical clipping. Over 400 discrete time steps, high-frequency spatial noise undergoes local diffusive smoothing governed by the D2Q9 viscosity $\nu = c_s^2(\tau - 1/2)$, followed by nonlinear amplification of intermediate wavelengths. Macroscopic density fields extracted at every time step confirm the gradual coalescence of isotropic snowy fluctuations into coherent, interconnected phase domains suitable for real-time visual synthesis.

## V. CONCLUSION AND FUTURE WORK

We have established a theoretical and numerical framework translating analog video feedback synthesis into a nonlinear D2Q9 lattice Boltzmann model. By embedding a sigmoidal saturation operator directly within a compile-time polymorphic collision kernel and bridging the core solver to an interactive Python GUI via `pybind11`, the prototype successfully captures the real-time competition between spatial diffusion and nonlinear gain amplification characteristic of spinodal-like pattern formation.

With the core real-time prototype operational, future development will focus on three unexplored directions:

1. **Multichannel RGB Color Dynamics:** Extending the single-variable density formulation to coupled vector fields to simulate chromatic aberration, hue separation, and RGB channel phase competition.
2. **Spatial Affine Transformations (Optical Warping):** Incorporating coordinate rotation, scaling, and translation operators within the streaming and forcing steps to replicate physical camera-to-CRT optical feedback geometry.
3. **GPU & Compute Shader Acceleration:** Offloading the lattice collision and streaming kernels to CUDA or compute shaders to support high-definition ($1080p$ to $4K$) real-time visual synthesis.


[1] J. W. Cahn and J. E. Hilliard, Free Energy of a Nonuniform System. I. Interfacial Free Energy, J. Chem. Phys. **28**, 258 (1958).

[2] S. Succi, *The Lattice Boltzmann Equation for Fluid Dynamics and Beyond* (Oxford University Press, Oxford, 2001).

[3] T. Krüger, H. Kusumaatmaja, A. Kuzmin, O. Shardt, G. Silva, and E. M. Viggen, *The Lattice Boltzmann Method: Principles and Practice* (Springer, Cham, 2017).